

Diary entry for Brown-Representability

Aditya Pahuja

2026-06-22

Contents

1	Ω-spectra represent cohomology theories	2
2	Cohomology theories are represented by Ω-spectra	2

1 Ω -spectra represent cohomology theories

Definition 1.1 (Ω -spectrum). An Ω -spectrum is a sequence of spaces $K_1, K_2, \dots$ equipped with (weak) homotopy equivalences $\xi_n: K_n \rightarrow \Omega K_{n+1}$ for each n .

Example 1.2 (Eilenberg-MacLane spectra). Let $K_n = K(G, n)$ be the n th Eilenberg-MacLane space. For a given n and any k ,

$$\pi_k(K_n) \cong \pi_{k+1}(K_{n+1}) = \pi_k(\Omega K_{n+1}),$$

so $K_n \simeq \Omega K_{n+1}$ because Eilenberg-MacLane spaces are unique up to homotopy. Choosing ξ_n to be a homotopy equivalence from K_n to ΩK_{n+1} thus gives an Ω -spectrum for each abelian group G , often denoted HG and called the **Eilenberg-MacLane spectrum of G** .

Theorem 1.3 (Ω -spectra represent cohomology theories). Let $\{(K_n, \xi_n)\}$ be an Ω -spectrum. The sequence of contravariant functors $H^n = [-, K_n]$ from the category of pointed CW complexes to the category of abelian groups satisfies the axioms for a cohomology theory, namely:

- (i) If $f \simeq g: X \rightarrow Y$, then $f^* = g^*: H^n(Y) \rightarrow H^n(X)$.
- (ii) For each CW pair (X, A) there is a long exact sequence

$$\dots \xrightarrow{\delta} H^n(X/A) \xrightarrow{q^*} H^n(X) \xrightarrow{i^*} H^n(A) \xrightarrow{\delta} H^{n+1}(X/A) \xrightarrow{q^*} \dots$$

Equivalently, $H^n(X) \cong H^{n+1}(\Sigma X)$ and $H^n(X/A) \rightarrow H^n(X) \rightarrow H^n(A)$ is exact for all X and n .

- (iii) Each H^n sends coproducts to products, i.e. for a wedge $X = \bigvee_{\alpha} X_{\alpha}$ with inclusions $i_{\alpha}: X_{\alpha} \rightarrow X$, the map

$$\prod_{\alpha} i_{\alpha}^*: H^n(X) \rightarrow \prod_{\alpha} H^n(X_{\alpha})$$

is an isomorphism.

Remark 1.4. The set $[X, K_n]$ has an (abelian) group structure for each X due to $K_n \simeq \Omega^2 K_{n+2}$, with the group operation $f + g$ being $(f + g)(x) = f(x) \cdot g(x)$, the composition of the two maps $S^2 \rightarrow K_{n+2}$ given by $f(x)$ and $g(x)$.

Proof. (i) If f and g are homotopic, then for any homotopy class $[\gamma] \in [Y, K_n]$,

$$f^*[\gamma] = [\gamma \circ f] = [\gamma \circ g] = g^*[\gamma].$$

(ii) TODO

(iii) The map into the product is invertible by sending a collection of maps $f_{\alpha}: X_{\alpha} \rightarrow K_n$ to the map which is equal to f_{α} on each copy of X_{α} inside X . $\square$

uses the cofibration sequence $A \rightarrow X \rightarrow X/A \rightarrow \Sigma A \rightarrow \dots$ or something

2 Cohomology theories are represented by Ω -spectra

The converse of Theorem 1.3 is also true — every cohomology theory is represented by an Ω -spectrum! To prove this, we first use a representability result for a single nice enough functor:

Theorem 2.1 (Brown representability theorem). Let $H: \mathbf{CW}^* \rightarrow \mathbf{Set}^*$ be a contravariant functor from the category of connected pointed CW complexes to the category of sets with the following properties:

- (i) (Homotopy invariance) If $f \simeq g: X \rightarrow Y$ then $H(f) \simeq H(g): H(Y) \rightarrow H(X)$.

(ii) (Mayer-Vietoris axiom) Suppose X is the union of (connected) subcomplexes A and B . If there are $a \in H(A)$ and $b \in H(B)$ such that $a|_{A \cap B} = b|_{A \cap B}$ then there exists $x \in H(X)$ such that $x|_A = a$ and $x|_B = b$ (where the restriction is the pullback under inclusion).

(iii) (Wedge axiom) For a wedge $X = \bigvee_{\alpha} X_{\alpha}$ with inclusions $i_{\alpha}: X_{\alpha} \rightarrow X$, the map

$$\prod_{\alpha} i_{\alpha}^*: H(X) \rightarrow \prod_{\alpha} H(X_{\alpha})$$

is an isomorphism.

Then, there exists $K \in \mathbf{CW}^*$ and $u \in H(K)$ such that $T_X: [X, K] \rightarrow H(X)$ given by $T_X(f) = f^*u$ is an isomorphism for all X , i.e. $[-, K]$ and H are naturally isomorphic.

why need connected?

Remark 2.2. The proper setting for this theorem should be the homotopy category of connected pointed CW complexes or something, in which case the functor is representable in the sense of being naturally isomorphic to a contravariant Hom functor (in $\mathbf{Set}^*$).

The idea of the proof is that we construct a pair (K, u) that satisfies the theorem statement when X is a sphere and then show that this turns out to also work for all spaces.

Definition 2.3 (n -universal, π_* -universal). A pair (K, u) with $K \in \mathbf{CW}^*$ and $u \in H(K)$ is **n -universal** if the transformation $T_{S^i}^u: \pi_i(K) = [S^i, K] \rightarrow H(S^i)$ defined in the statement of Theorem 2.1 is an isomorphism for $i < n$ and surjective for $i = n$. The pair is **π_* -universal** if it is n -universal for all n , i.e. $T_{S^i}^u$ is an isomorphism for all i .

Remark 2.4. The word “universal” is because of the following: If (K, u) and (K', u') are π_* -universal and $f: K \rightarrow K'$ is a map such that $f^*u' = u$, then $T_{S^n}^{u'} \circ f_* = T_{S^n}^u$ where $f_*: \pi_n(K) \rightarrow \pi_n(K')$ is the induced homomorphism on homotopy groups, so f is a bijection, hence an isomorphism. This implies that f is a weak homotopy equivalence, which we can upgrade to a homotopy equivalence because K and K' are CW complexes.

Lemma 2.5. For any pair (Z, z) with $Z \in \mathbf{CW}^*$ and $z \in H(Z)$, there exists a π_* -universal pair (K, u) having Z as a subcomplex and $u|_Z = z$.

Proof. Let $K_1 = Z \vee \bigvee_{\alpha} S_{\alpha}^1$ over all $\alpha \in H(S^1)$. By the wedge axiom, there exists $u_1 \in H(K_1)$ such that $u_1|_Z = u$ and $u_1|_{S_{\alpha}^1} = \alpha$, so (K_1, u_1) is 1-universal.

The idea in general, given n -universal (K_n, u_n) , is to take maps $f_{\alpha}: S^n \rightarrow K_n$, one representing each element of the kernel of $T_{S^n}^{u_n}$, and form the mapping cone for $\bigvee_{\alpha} f_{\alpha}$ to kill the kernel elements, and then to wedge with one $(n+1)$ -sphere for each element of $H(S^{n+1})$ to get surjectivity. Then, to pass to the limit (to get n -universality for all n), form the mapping telescope of the K_n s. $\square$

details

Lemma 2.6. Let (K, u) be a π_* -universal pair and let (X, A) be a pointed CW pair. Suppose $x \in H(X)$ and that $f: A \rightarrow K$ with $f^*u = x|_A$. Then f extends to a map $g: X \rightarrow K$ with $g^*u = x$.

Proof. Construct the reduced mapping cylinder of f ,

$$M_f = K \cup (A \times [0, 1]) / (f(a) \sim (a, 0), (x_0, t) \sim (k_0));$$

where x_0 and k_0 are the basepoints of X and K ; since $M_f \simeq K$ we can regard u as an element of $H(M_f)$ (more precisely, there is a $\tilde{u} \in H(M_f)$ which restricts to u) whose restriction to $A \times \{1\}$ is equal to $x|_A$, since the inclusion sending A to $A \times \{1\}$ is homotopic to f . This means we can work with the inclusion $a \mapsto (a, 1)$ of A into M_f instead of with f .

Then, attach X to M_f along $A \times \{1\}$ to form Z . By the Mayer-Vietoris axiom there exists $z \in H(Z)$ which restricts to x and to u , and by Lemma 2.5 there is a π_* -universal pair (K', u') with Z as a subcomplex of K' such that u' restricts to z . The argument of

Remark 2.4 now tells us that $K' \simeq M_f$, so M_f deformation retracts onto M_f , inducing a homotopy from the inclusion of X into K' to a map $g: X \rightarrow M_f$. This homotopy is constant on A , so g extends the inclusion of A into M_f , and $g^*u = g^*j^*u' = u'|_X = x$ where $j: M_f \hookrightarrow K'$, by homotopy invariance of H . $\square$

Proof of Theorem 2.1. We will show that a π_* -universal pair (K, u) does the job. Taking A to be a point in Lemma 2.6, note that $H(A)$ consists of a single point, say by applying the wedge axiom to $B \vee A$ for any space B with $H(B)$ nontrivial; this means $f^*u = x|_A$ holds for all $x \in H(X)$, so for each $x \in H(X)$ there is some $g_x: X \rightarrow K$ with $g_x^*u = x$.

Now, suppose that $T_X^u[f_0] = T_X^u[f_1]$ for some $f_0, f_1: X \rightarrow K$ and let $p: X \times I \rightarrow X$ be the projection (where $X \times I$ is the reduced cylinder, with $\{x_0\} \times I$ collapsed to a point). Define x to be $p^*f_0^*u = p^*f_1^*u$; since projection is a left inverse to the inclusions of X into the cylinder, $x|_{X \times \{0\}}$ and $x|_{X \times \{1\}}$ are both equal to $f_0^*u = f_1^*u$. Therefore $x|_{(X \times \{0\}) \vee (X \times \{1\})}$ is equal to (f_0^*u, f_1^*u) . By Lemma 2.6, $f = f_0 \vee f_1: (X \times \{0\}) \vee (X \times \{1\}) \rightarrow K$ extends to $X \times I$, so f_0 and f_1 are homotopic, proving that T_X^u is injective. $\square$